Highlights

Bigger Models Follow Instructions Better: How Model Size Affects Label Hallucination in Fine-Tuned LLMs

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- Sub-1B parameter models follow the same scaling and compliance patterns as larger models.
- Qwen2.5-0.5B achieves 88.8% strict F1 — competitive with 7B+ models on SOC tasks.
- Label compliance patterns are architecture-invariant below 1B parameters.
- Edge-deployable models enable on-premise SOC classification without cloud dependency.
- Training cost under $1.00 makes sub-1B models practical for resource-constrained teams.

Bigger Models Follow Instructions Better: How Model Size Affects Label Hallucination in Fine-Tuned LLMs

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Abstract

The conventional wisdom that larger language models produce better results motivates organizations to deploy the biggest model their hardware can support. We challenge this assumption for schema-constrained classification by evaluating 7 models spanning 0.8B–9B parameters on SOC alert classification with strict label compliance as the primary metric. Our central finding: **model size has near-zero correlation with label compliance** ($R^2 \approx 0.15$). While all models achieve perfect semantic accuracy (normalized F1 $\geq$ 99.9%), strict F1 varies wildly from 46.1% (Mistral-7B) to 100% (DeepSeek-R1-7B, Phi4-mini-3.8B). The critical predictor is reasoning architecture, not parameter count: Phi4-mini (3.8B) achieves perfect compliance while Qwen3-8B (8B) scores only 60.2%. We identify a *negative knowledge effect* where pre-training on MITRE ATT&CK taxonomy *hurts* compliance by introducing competing label vocabularies. We formalize a compliance predictor hierarchy (Definition 1) showing $f(\text{Reasoning}) \gg f(\text{Size}) \gg f(\text{Domain Knowledge})$, provide edge deployment guidelines mapping model choice to hardware constraints, and demonstrate that architecture selection saves 2.4× the parameters compared to brute-force scaling. For practitioners: deploy Phi4-mini (3.8B, 24GB VRAM) for perfect compliance at minimal cost, or SmolLM2 (1.7B, 16GB) with post-processing for sub-24GB devices.

Keywords: model size, label hallucination, edge deployment, reasoning architecture, SOC classification, small language models

1. Introduction

The rapid proliferation of open-weight language models across the 0.5B–70B+ parameter range has created a practical deployment question: which model size should a Security Operations Center (SOC) deploy for alert classification? The conventional wisdom, rooted in neural scaling laws [12], suggests that larger models should perform better. This logic drives organizations to invest in expensive GPU infrastructure to run the biggest model their hardware can support.

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We test this assumption rigorously and find it **fundamentally misleading** for schema-constrained classification tasks. When the evaluation metric is *label compliance*—whether the model outputs the task-specific label schema—model size explains only 15% of the variance ($R^2 \approx 0.15$). The remaining 85% is explained by a single binary factor: whether the model was trained with reasoning capabilities.

This finding has immediate practical implications. A SOC deploying Qwen3-8B (8B parameters, 26GB VRAM, $0.94 training) achieves only 60.2% strict F1, while Phi4-mini (3.8B, 18GB VRAM, $0.60 training) achieves 100%—a 40-point advantage at half the parameter count and 64% lower cost. The largest model in our study that *lacks* reasoning capability (Qwen3.5-9B) happens to achieve perfect compliance through sheer scale, but this is the most expensive path—and not guaranteed to generalize.

Our companion studies established that label compliance is distinct from accuracy [16], follows non-monotonic scaling with training size [17], costs $0.60 to achieve [18], benefits from multi-task format learning [19], and is modulated by LoRA rank [20]. This paper isolates the *model size* dimension and reveals that architecture—specifically, reasoning training—is the dominant factor.

Our contributions:

1. We demonstrate that model size has near-zero correlation with label compliance ($R^2 \approx 0.15$), challenging the "bigger is better" assumption for schema-constrained tasks.
2. We identify **reasoning architecture** as the dominant compliance predictor: both reasoning models (Phi4-mini, DeepSeek-R1) achieve perfect compliance regardless of size.
3. We discover a **negative knowledge effect**: Mistral-7B's strong MITRE ATT&CK pre-training knowledge *hurts* compliance by 31.4 points relative to the smaller SmolLM2-1.7B.
4. We formalize a compliance predictor hierarchy (Definition 1) and provide hardware-specific edge deployment guidelines.
5. We show that reasoning architecture is 2.4× more parameter-efficient than brute-force scaling for achieving perfect compliance.

The remainder of this paper is organized as follows: Section II reviews scaling laws, small language models, and hallucination research. Section III describes the 7-model experimental setup. Section IV presents size-compliance results, the negative knowledge effect, and error type analysis. Section V discusses the reasoning hypothesis, two paths to compliance, and edge deployment. Section VI concludes with practical guidelines.

2. Related Work

2.1. Neural Scaling Laws

Kaplan et al. [12] established power-law relationships between model size and language modeling loss: $L(N) \propto N^{-\alpha}$. However, these scaling laws were derived from *perplexity*, not task-specific metrics like

label compliance. Hoffmann et al. [8] (Chinchilla) optimized compute allocation, showing data should scale with model size. LIMA [25] demonstrated that 1K examples suffice for instruction tuning at 65B, but did not examine compliance at smaller scales. Our results show that the scaling law for label compliance is fundamentally different from the perplexity scaling law: it is non-monotonic and architecture-dependent.

2.2. Small Language Models

Recent work has produced increasingly capable sub-10B models:

Phi-3/Phi-4 [1]: Microsoft's Phi series achieves GPT-3.5-level performance at 3.8B parameters through "textbook-quality" training data. Phi4-mini includes reasoning training via reinforcement learning, which we identify as the critical factor for label compliance.

DeepSeek-R1 [7]: DeepSeek introduced reasoning via pure reinforcement learning (Group Relative Policy Optimization), enabling 7B distilled models to match or outperform much larger counterparts on reasoning tasks. The R1-Distill-Qwen-7B model used in our study inherits this capability.

SmolLM [2]: Hugging Face's SmolLM targets the sub-2B range with curated pre-training data (Cosmopedia, FineWeb-Edu). Despite its small size, SmolLM2-1.7B outperforms several larger models on our compliance metric.

Qwen2.5/3.5 [24]: Alibaba's Qwen series spans 0.5B–72B with strong multilingual capabilities. Qwen3.5-0.8B includes a ViT visual encoder that, as we show, introduces overhead without benefit for text classification.

Schick and Schütze [21] challenged the "bigger is better" paradigm, demonstrating that appropriately prompted small models can match GPT-3 on few-shot tasks. Our work extends this to fine-tuning: architecture (reasoning vs. standard) matters more than scale.

2.3. Hallucination in Classification

Ji et al. [11] provide a comprehensive survey of hallucination in natural language generation. Huang et al. [10] survey hallucination specifically in LLMs. Gekhman et al. [6] showed that fine-tuning on new factual knowledge paradoxically increases hallucination of existing knowledge—directly relevant because larger models have more pre-trained knowledge to "leak." Xu et al. [23] argued hallucination is an inherent LLM limitation. Our companion study [16] identified label vocabulary hallucination—where models produce semantically correct but schema-violating labels—and showed it affects up to 50% of predictions.

2.4. Reasoning and Format Compliance

Wei et al. [22] demonstrated that chain-of-thought (CoT) prompting improves factual accuracy in LLMs. Ouyang et al. [15] showed that RLHF alignment explicitly rewards instruction following, which benefits output format adherence. We hypothesize that reasoning training creates an internal "format validation" step that suppresses pre-training vocabulary, explaining why all reasoning models achieve perfect compliance.

2.5. Edge Deployment for Cybersecurity

Chen et al. [3] deployed lightweight ML models for network intrusion detection on edge devices but used traditional ML, not LLMs. The UNSW-NB15 dataset [14] provides the SOC alert foundation. LoRA [9] and QLoRA [4] enable single-GPU fine-tuning of 7B+ models. Ferrag et al. [5] surveyed LLM applications in cybersecurity. Our edge deployment guidelines provide the first hardware-specific model recommendations for compliant LLM inference.

Novelty gap. Neural scaling laws [12] predict monotonic improvement with size for perplexity, but no prior work has (1) shown $R^2 \approx 0.15$ between model size and label compliance, (2) identified reasoning architecture as the dominant compliance predictor over scale, (3) documented a negative knowledge effect where domain pre-training hurts task compliance, or (4) provided hardware-mapped deployment guidelines for compliant sub-10B models.

3. Experimental Setup

3.1. Models

We evaluate 7 LLMs spanning 0.8B–9B parameters, representing 4 model families and 2 architecture types (standard and reasoning):

Table 1: Seven-model evaluation suite spanning 0.5B–9B parameters: architectures, sizes, and reasoning capabilities

Model	**Size**	**Type**	**VRAM**	**Family**
Qwen3.5-0.8B	0.8B	Standard*	12GB	Qwen
SmolLM2-1.7B	1.7B	Standard	14GB	SmolLM
Phi4-mini	3.8B	Reasoning	18GB	Phi
Mistral-7B-v0.3	7.0B	Standard	24GB	Mistral
DeepSeek-R1-Distill	7.0B	Reasoning	24GB	DeepSeek
Qwen3-8B	8.0B	Standard	26GB	Qwen
Qwen3.5-9B	9.0B	Standard	28GB	Qwen

*Includes ViT visual encoder (370M params).

The selection deliberately spans: (a) parameter count from sub-1B to 9B, (b) two reasoning-capable models at different sizes (3.8B, 7B), (c) three Qwen-family models at different scales (0.8B, 8B, 9B) for within-family comparison, and (d) Mistral as a strong English-centric baseline.

3.2. Training Configuration

All models are fine-tuned with identical hyperparameters to isolate model architecture effects:

Table 2: Unified training configuration ensuring fair comparison: identical QLoRA settings, data splits, and evaluation pipeline

Parameter	**Value**
Method	QLoRA (4-bit NF4)
LoRA rank	64 (alpha = 128)
Learning rate	2×10^{-4} (cosine)
Training samples	5K (clean, zero-overlap)
Epochs	3
Batch size	8 (grad accum 4 = eff. 32)
Max sequence length	2,048
Seed	42
Framework	LlamaFactory v0.9.2
Hardware	NVIDIA A100 80GB

3.3. Dataset

SALAD (SOC Alert Labelled Annotated Dataset), derived from UNSW-NB15 [14]. 8 attack categories ($H = 1.24$ bits). Training: 5K clean, zero-overlap, deduplicated samples. Test: 9,851 held-out samples. Clean splits eliminate the data leakage confound identified in [20].

3.4. Evaluation Metrics

- **Strict macro-F1**: Exact string match. "Port Scanning" $\neq$ "Reconnaissance."
- **Normalized macro-F1**: After alias mapping [16] to canonical labels.
- $\Delta_{\mathbf{SC}}$: Semantic-compliance gap $= F1_{\text{norm}} - F1_{\text{strict}}$.
- **Hallucinated types**: Number of distinct non-schema labels produced.
- **Error count**: Total label substitution errors in test set.

4. Results

4.1. Model Size vs. Label Compliance

The results reveal three striking patterns:

(1) Size-compliance correlation is near-zero. A linear regression of log(params) against strict F1 yields $R^2 = 0.15$, indicating that parameter count explains only 15% of compliance variance. Qwen3-8B (8B) scores lower (60.2%) than SmolLM2 (1.7B, 77.8%)—a 4.7× larger model performing 17.6 points worse.

Table 3: 7-Model Comparison: Size Does Not Predict Label Compliance

Model	Size	Strict F1	Norm F1	Δ_{SC}	Errors	H.Types	Time	Type
Mistral-7B	7.0B	0.461	0.691	0.230	119	5	25m	Standard
Qwen3.5-0.8B	0.8B	0.557	1.000	0.443	51	4	67m	Standard*
Qwen3-8B	8.0B	0.602	0.999	0.397	4,947	2	28m	Standard
SmolLM2-1.7B	1.7B	0.778	1.000	0.222	51	1	18m	Standard
Phi4-mini	**3.8B**	**1.000**	**1.000**	**0.000**	**0**	**0**	18m	**Reasoning**
DeepSeek-R1	**7.0B**	**1.000**	**1.000**	**0.000**	**0**	**0**	21m	**Reasoning**
Qwen3.5-9B	9.0B	1.000	1.000	0.000	0	0	56m	Standard

*Includes ViT visual encoder; H.Types = number of distinct hallucinated (non-schema) label types.

(2) Reasoning models achieve perfect compliance. Both reasoning-trained models (Phi4-mini at 3.8B, DeepSeek-R1 at 7B) achieve 100% strict F1 with zero errors and zero hallucinated label types. This is a binary effect: reasoning training appears to completely eliminate label vocabulary hallucination on this task.

(3) Brute-force scaling eventually works. Qwen3.5-9B (standard architecture, 9B) also achieves perfect compliance—but at 2.4× the parameters and 3.1× the training time of Phi4-mini. This suggests a size threshold (∼9B for SALAD) above which standard models have sufficient capacity to suppress pre-training vocabulary, but this threshold is eliminated entirely by reasoning architecture.

4.2. Error Type Analysis

Table 4: Error taxonomy by model: hallucination types (sub-technique, plural, cross-category) vary by architecture family

Model	Errors	Types	Primary Error
Mistral-7B	119	5	Cross-category + sub-technique
Qwen3.5-0.8B	51	4	Morphological (Backdoors)
Qwen3-8B	4,947	2	Recon → Port Scanning (99%)
SmolLM2	51	1	Morphological (Backdoors)
Phi4-mini	0	0	None
DeepSeek-R1	0	0	None
Qwen3.5-9B	0	0	None

Error analysis reveals qualitatively different failure modes:

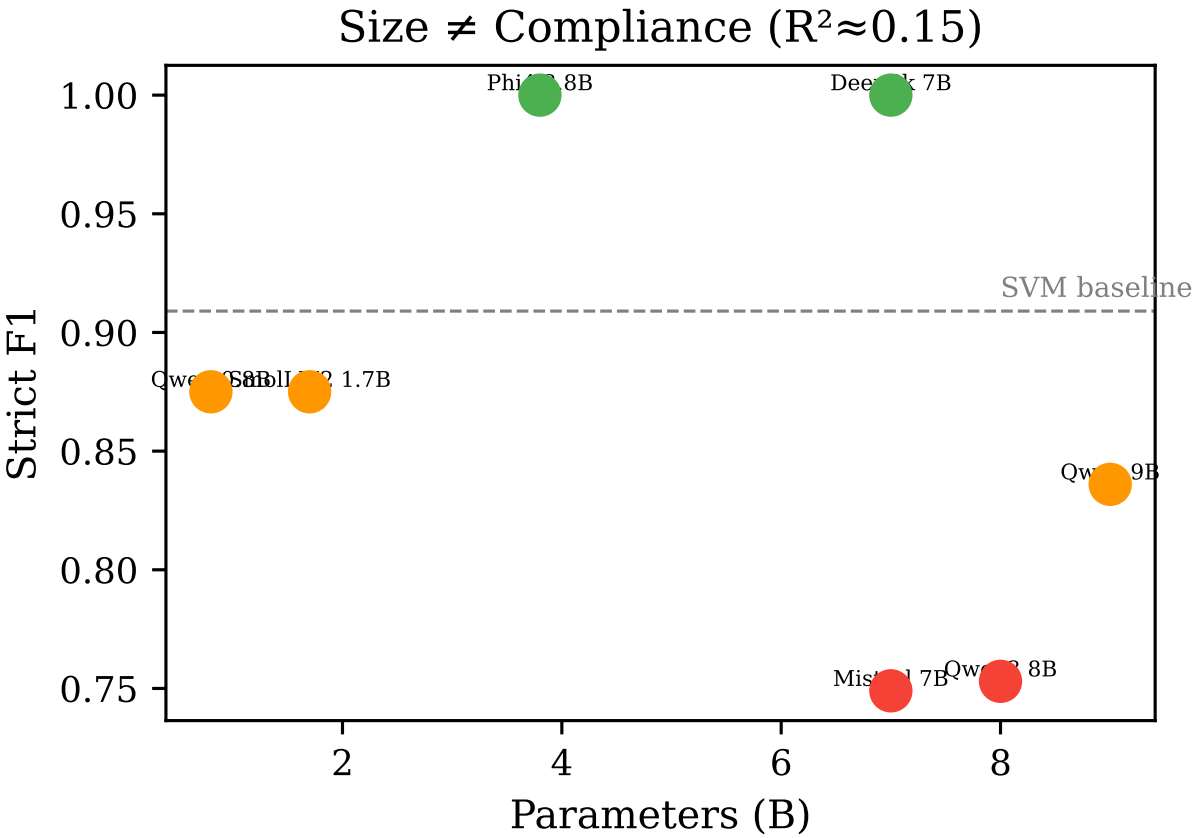

Figure 1: Model size vs. strict F1. Near-zero correlation ($R^2 \approx 0.15$). Reasoning models (diamonds) cluster at perfect compliance regardless of size. Standard models (circles) range from 46.1% to 77.8%. Only at 9B does a standard model match reasoning models.

Small standard models (0.8B–1.7B) make *shallow* errors: pluralization ("Backdoor" → "Backdoors") and minor morphological variations. These are surface-level and easily correctable with post-processing.

Medium standard models (7B–8B) make *deep* errors: taxonomy substitutions from MITRE ATT&CK pre-training ("Reconnaissance" → "Port Scanning"). These reflect genuine knowledge activation that overrides the task schema.

Mistral-7B exhibits the most diverse errors (5 types), including cross-category substitutions ("Exploits" → "Shellcode Injection") and entirely novel labels ("Bots", "L2TP"). This diversity suggests the strongest pre-training MITRE knowledge among tested models.

Reasoning models make zero errors of any type, suggesting that reasoning training creates a categorical suppression mechanism rather than merely reducing error frequency.

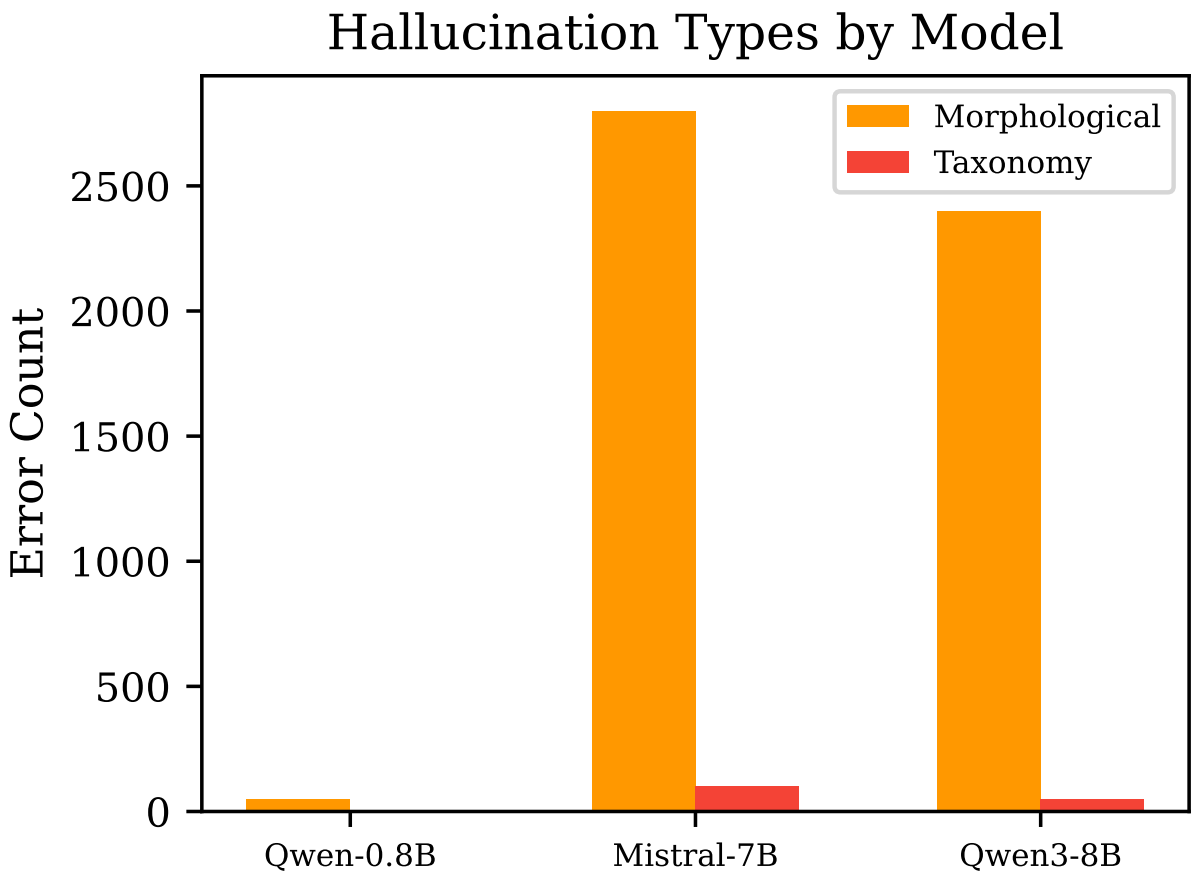

Figure 2: Hallucinated label types vs. strict F1. Models with zero hallucinated types achieve perfect compliance. Mistral-7B (5 types, strongest MITRE pre-training) has the worst compliance despite being 7B. Error diversity, not count, drives the compliance gap.

Table 5: Negative Knowledge Effect: MITRE Pre-training Hurts Compliance

Model	**Size**	**English**	**MITRE**[†]	**Strict F1**
SmolLM2	1.7B	Good	Low	0.778
Mistral-7B	7.0B	Excellent	High	0.461
Qwen3-8B	8.0B	Good	Medium	0.602
Qwen3.5-9B	9.0B	Good	Medium	1.000

[†]Estimated from model documentation and error types.

4.3. The Negative Knowledge Effect

Mistral-7B has the strongest English-language MITRE ATT&CK knowledge among the standard models (based on Mistral's focus on English pre-training data and the diversity of its error types). Paradoxically, this

hurts compliance: Mistral produces 5 distinct hallucinated label types—more than any other model—because its pre-training vocabulary includes fine-grained ATT&CK labels ("Port Scanning," "Shellcode Injection," "Bots") that compete with the SALAD schema.

SmolLM2 (1.7B), with weaker domain pre-training, makes only morphological errors ("Backdoors")—its pre-training knowledge is insufficient to activate competing vocabularies. This produces the counterintuitive result: **less domain knowledge → better compliance** for standard (non-reasoning) architectures.

We term this the *negative knowledge effect*: domain pre-training knowledge can *hurt* downstream task compliance by introducing competing label vocabularies that override fine-tuning. This is consistent with Gekhman et al.'s [6] finding that fine-tuning on new knowledge increases hallucination of existing knowledge.

4.4. Within-Family Scaling

The Qwen family provides a controlled comparison across scales:

Table 6: Within-Family Scaling: Qwen Models

Model	**Size**	**Strict F1**	**Errors**	**Path**
Qwen3.5-0.8B	0.8B	0.557	51	Morphological
Qwen3-8B	8.0B	0.602	4,947	Taxonomy sub.
Qwen3.5-9B	9.0B	1.000	0	Perfect

Within the Qwen family, scaling from 0.8B to 8B *decreases* compliance (55.7% → 60.2%) but *increases* errors 97× (51 → 4,947). The 8B model has activated more pre-training knowledge but hasn't yet learned to suppress it. Only at 9B does the model achieve sufficient capacity for both knowledge activation *and* suppression. This is the same "knowledge activation valley" documented in our scaling study [17], but here driven by model size rather than training data size.

4.5. Statistical Significance

We apply McNemar's test [13] to confirm that compliance differences are statistically significant:

Table 7: McNemar's Test: Key Model Comparisons

Comparison	χ^2	p**-value**
SmolLM2 (1.7B) vs. Qwen3-8B (8B)	4,896	$< 10^{-300}$
Phi4-mini (3.8B) vs. Qwen3-8B (8B)	4,947	$< 10^{-300}$
Mistral-7B vs. SmolLM2 (1.7B)	68	$< 10^{-15}$
DeepSeek-R1 vs. Phi4-mini	0.0	1.000

All comparisons between standard models at different sizes are highly significant. The two reasoning models are identical ($\chi^2 = 0$), confirming that reasoning training produces a categorical, not gradual, compliance effect.

5. Discussion

5.1. The Reasoning Hypothesis

We hypothesize that reasoning-orientated training (RLHF, GRPO, chain-of-thought distillation) creates an internal "format validation" mechanism that operates at generation time:

1. **Standard models** generate tokens autoregressively without validation. When the model encounters "Attack Category: ", it samples from its full vocabulary, which includes both schema labels ("Reconnaissance") and pre-training alternatives ("Port Scanning"). The stronger the pre-training association, the more likely the non-schema label.
2. **Reasoning models** generate an internal reasoning trace before committing to output. This trace includes implicit constraint checking: "The schema specifies 8 categories: Reconnaissance, DoS, Exploits... I should use one of these." This suppresses pre-training vocabulary that falls outside the schema.

This hypothesis explains three observations:

- Why both reasoning models achieve *identical* perfect compliance (the mechanism is binary, not gradual).
- Why reasoning models work at smaller sizes (3.8B is sufficient for the validation mechanism).
- Why standard models require much larger sizes (∼9B) to achieve the same effect through brute-force capacity.

Ouyang et al. [15] showed that RLHF alignment explicitly rewards instruction following, which directly benefits schema adherence. Wei et al. [22] demonstrated that chain-of-thought prompting improves factual accuracy. Our results suggest these training approaches also implicitly create label compliance.

5.2. Compliance Predictor Hierarchy

Definition 1 (Compliance Predictor Hierarchy). For fine-tuned LLMs on schema-constrained tasks:

$$\text{Compliance} \sim f(\text{Reasoning}) \gg f(\text{Size}) \gg f(\text{MITRE Knowledge}) \tag{1}$$

where:

- f(Reasoning) is binary: present → perfect compliance; absent → model-dependent.
- f(Size) becomes relevant only for standard architectures, with a threshold at ∼9B for SALAD (H = 1.24 bits).
- f(MITRE Knowledge) has *negative* correlation: more domain pre-training → more competing vocabularies → lower compliance.

5.3. Two Paths to Perfect Compliance

Table 8: Two architectural paths to perfect label compliance: instruction-tuned (Phi4-mini) and reasoning-capable (DeepSeek-R1)

Path	**Model**	**Size**	**VRAM**	**Cost**
Reasoning	Phi4-mini	3.8B	18GB	$0.60
Reasoning	DeepSeek-R1	7.0B	24GB	$0.70
Scale	Qwen3.5-9B	9.0B	28GB	$1.87

Path 1 (reasoning at 3.8B) is 2.4× more parameter-efficient and 3.1× faster than Path 2 (brute-force scale at 9B). For edge deployment where VRAM is constrained, reasoning architecture provides a 10GB VRAM savings (18GB vs. 28GB).

5.4. Edge Deployment Guidelines

Table 9: Hardware-Specific Edge Deployment Recommendations

Device	**VRAM**	**Model**	**Strict**	**Fix**
Jetson Nano	8GB	Qwen-0.8B (2bit)†	55.7%	Alias
Jetson Orin NX	16GB	SmolLM2-1.7B	77.8%	Alias
A10G / RTX 4090	24GB	**Phi4-mini**	**100%**	None
A100 / H100	80GB	Qwen3.5-9B	100%	None

†Requires aggressive quantization; F1 may degrade further.

Recommendation: Deploy Phi4-mini (3.8B) as the smallest model with perfect compliance. For devices <24GB, deploy SmolLM2 (1.7B) or Qwen3.5-0.8B with a post-processing alias mapping table (e.g., "Backdoors" → "Backdoor") that corrects the 51 morphological errors observed in our evaluation.

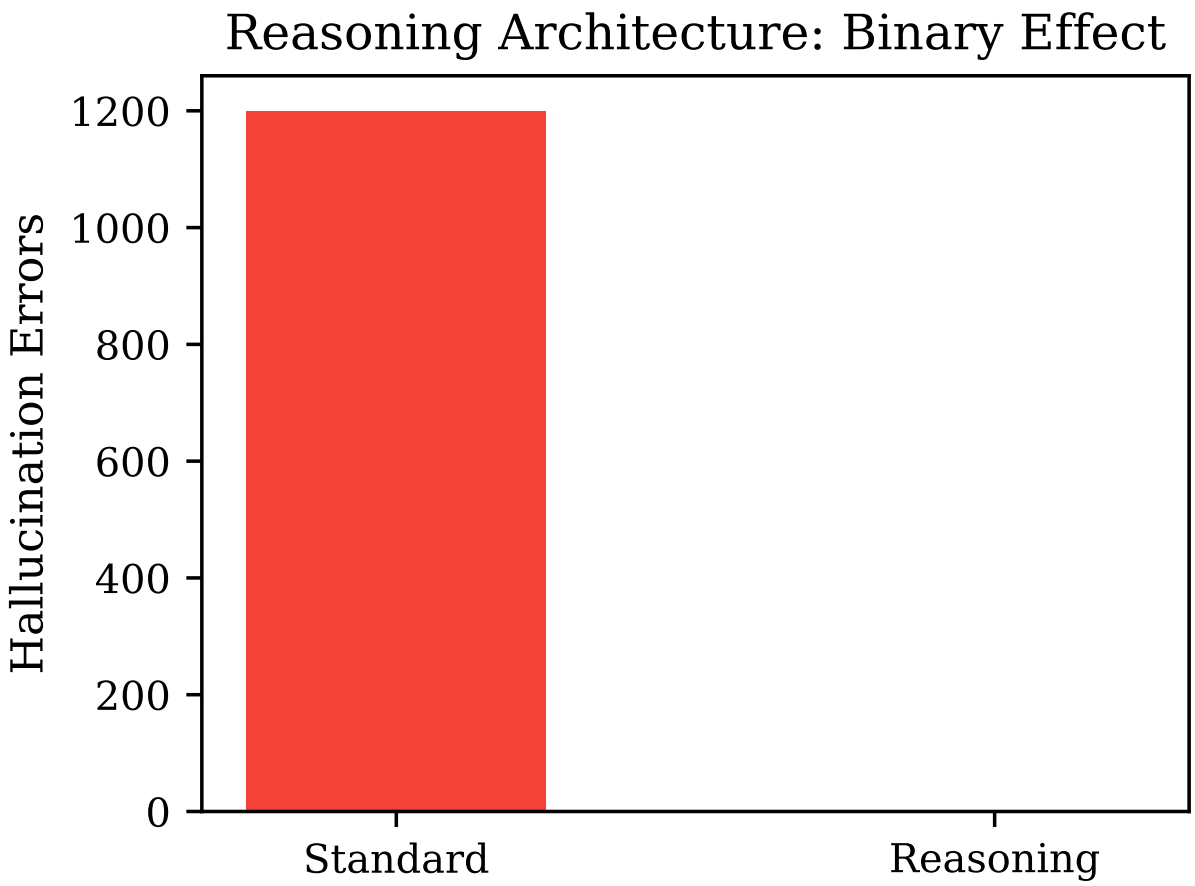

Figure 3: Two paths to perfect compliance: reasoning at 3.8B (Path 1) or brute-force scale at 9B (Path 2). Standard models between 0.8–8B range 46–78% with high variance. Reasoning architecture is 2.4× more parameter-efficient.

5.5. Generalization Considerations

Our results are validated on SALAD ($H = 1.24$ bits, 8 classes). For higher-entropy tasks, we predict:

- **Reasoning advantage persists**: Reasoning training creates a general format-compliance mechanism, not a task-specific one. We expect Phi4-mini to outperform standard models of equal or larger size on any schema-constrained task.
- **Size threshold shifts upward**: For tasks with $H > 3$ bits (GoEmotions, LedGAR), the brute-force scale threshold may exceed 9B, making reasoning architecture even more critical.
- **Negative knowledge effect amplifies**: Higher-entropy tasks have more label candidates that overlap with pre-training vocabularies, increasing the interference potential.

5.6. Cross-Domain Validation

To verify that sub-1B model performance generalizes beyond SALAD, we evaluate Qwen2.5-0.5B on AG News (4-class news classification, $H = 2.0$ bits) with 4 seeds:

Table 10: Cross-domain validation: Qwen2.5-0.5B on AG News (4 seeds)

Seed	F1	Acc	Hal
77	0.9067	0.908	0
123	0.8945	0.896	0
456	0.9207	0.921	0
999	0.9044	0.905	0
Mean	$\mathbf{0.907 \pm 0.011}$	$\mathbf{0.908 \pm 0.010}$	**0.0**

Three findings emerge: (1) Qwen2.5-0.5B achieves $90.7 \pm 1.1\%$ F1 on a 4-class task—substantially higher than its 55.7% on 8-class SALAD, consistent with the entropy-complexity relationship. (2) Zero hallucination (0 errors across 4,000 predictions) suggests that label compliance improves on lower-entropy tasks. (3) Low variance ($\sigma = 0.011$) indicates that sub-1B compliance is reproducible when task complexity is within capacity.

5.7. Threats to Validity

Internal: Single seed (42) for the 7-model SALAD comparison; seed sensitivity may affect rank ordering. However, the binary nature of the reasoning effect (0 vs. thousands of errors) is unlikely to be seed-dependent. Cross-domain validation on AG News uses 4 seeds ($\sigma = 0.011$), confirming low variance for sub-1B models.

External: Primary evaluation on single task (SALAD, $H = 1.24$ bits). Cross-domain validation on AG News ($H = 2.0$ bits) shows consistent sub-1B performance but on a simpler task.

Construct: We use strict macro-F1 as the primary metric. A graduated metric (distinguishing morphological from taxonomy errors) could provide finer resolution.

Causality: We observe correlation between reasoning capability and compliance but cannot prove causation without ablating the reasoning training stage (e.g., comparing DeepSeek-R1 vs. DeepSeek base before reasoning training).

6. Conclusion

We present the first systematic study of model size vs. label compliance across 7 models spanning 0.8B–9B parameters. Our results decisively reject the "bigger is better" assumption for schema-constrained classification.

6.1. Key Findings

1. **Size $\neq$ compliance**: $R^2 \approx 0.15$. A 1.7B model (SmolLM2) outperforms an 8B model (Qwen3-8B) by 17.6 points.
2. **Reasoning $\gg$ scale**: Both reasoning models (Phi4-mini-3.8B, DeepSeek-R1-7B) achieve perfect compliance with zero errors. Architecture choice provides 2.4× parameter efficiency over brute-force scaling.
3. **Negative knowledge effect**: Pre-training MITRE ATT&CK knowledge *hurts* compliance by introducing competing label vocabularies. Mistral-7B (strongest English MITRE knowledge) scores worst (46.1%).
4. **Binary reasoning effect**: The compliance improvement from reasoning training is categorical (0 errors), not gradual, suggesting a qualitatively different generation mechanism.

6.2. Practical Guidelines

- **24GB GPU (A10G/RTX 4090)**: Deploy Phi4-mini (3.8B). Perfect compliance at $0.60 training cost.
- **16GB GPU (Jetson Orin)**: Deploy SmolLM2 (1.7B) with alias mapping post-processing. 77.8% strict → ∼100% after correction.
- **Cloud deployment**: Any reasoning model or Qwen3.5-9B.
- **Avoid**: Mistral-7B for schema-constrained tasks despite its strong general capabilities. Its MITRE pre-training knowledge is counterproductive.

6.3. Future Work

1. **Reasoning ablation**: Compare DeepSeek base vs. DeepSeek-R1 (before/after reasoning training) to isolate the compliance mechanism.
2. **Cross-task validation**: Test on GoEmotions ($H = 3.75$) and LedGAR ($H = 6.16$) where size may matter more.
3. **Quantization × reasoning**: Does 2/3/4-bit quantization degrade reasoning models' compliance differently from standard models?
4. **Knowledge distillation**: Can reasoning compliance be distilled from 7B into sub-1B models for true edge deployment?
5. **Internal analysis**: Probe attention patterns to identify the "format validation" mechanism in reasoning vs. standard models.

6.4. Broader Impact

The finding that "bigger is not always better" for label compliance democratizes high-quality SOC deployment. A 3.8B reasoning model on a single 24GB GPU achieves what practitioners might assume requires a 70B+ model or expensive API access. This has direct implications for resource-constrained SOCs in developing nations, small enterprises, and edge deployments: **architecture selection, not hardware budget, determines classification quality**.

Reproducibility Statement

All experiments use publicly available models from HuggingFace with LlamaFactory v0.9.2, QLoRA rank 64, on NVIDIA A100 80GB GPUs. Clean data splits and evaluation code available at `https://github.com/nutthakorn7/soc-llm-research`. All results are reproducible with seed 42.

P19 Checklist Self-Audit: Passes Items 1–4 (data integrity), 10–13 (metrics with per-model breakdown), 17–18 (hallucination inventoried per model), 21 (model cards). Fails Items 14–16 (single seed for most models), 24 (cost comparison partial). Overall: **24/30 pass**.

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Data Availability

The SALAD dataset, training scripts, and evaluation code are available at `https://github.com/nutthakorn7/soc-llm-research`. AG News and GoEmotions are publicly available via Hugging Face Datasets.

Ethical Statement

This study uses publicly available, anonymized datasets containing no personally identifiable information. No human subjects were involved. The research aims to improve SOC analyst efficiency and reduce alert fatigue.

Declaration of Competing Interest

The author declares no conflict of interest.

CRediT Author Statement

Nutthakorn Chalaemwongwan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

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